



**SOPHIE'S
BIONUTRIENTS**

Precision Fermentation & Sustainable Microalgae Protein

Maximum Protein,
Minimum Footprint

Eugene Wang

Co-founder & CEO



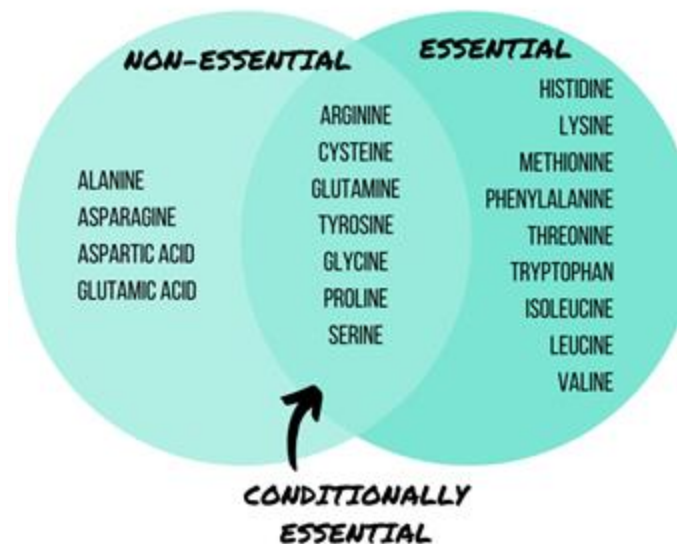
Our algae delivers calcium, iron, & essential amino acids



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AMINO ACIDS



NO Known Allergens

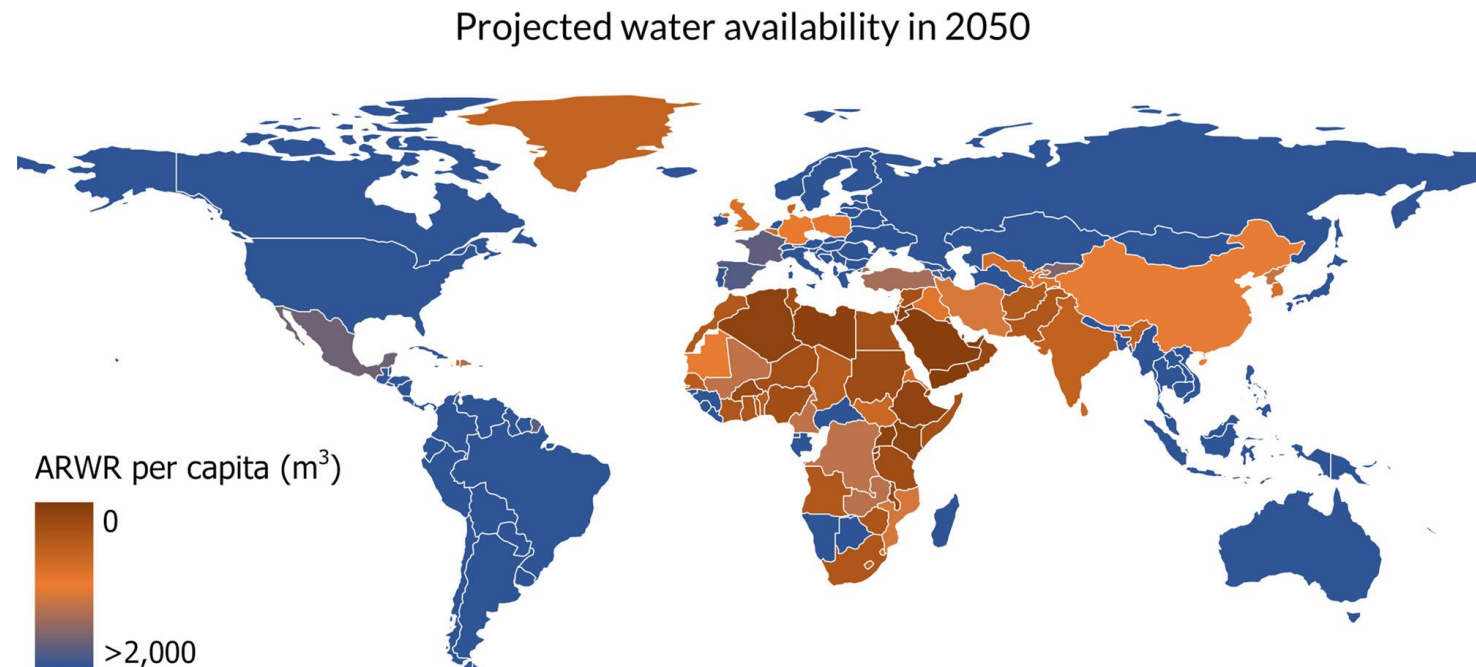
Regulatory friendly & Non-GMO – *Chlorella vulgaris*

Chlorella vulgaris - is widely recognized as safe (GRAS) by the FDA and is not considered a "novel food" in the EU, provided it has a history of consumption before May 1997.

Save time on regulatory approvals!!

Can be executed in Fresh Water, Brackish Water. and Seawater

Freshwater reserves are declining globally by roughly 3% annually, driven by unsustainable consumption, drought, and climate change, with over 75% of the world's population living in countries losing water.



Can be executed in any container (e.g. beer brewery) with proper aeration and stirring – Lower CAPEX & OPEX



Why Food Waste?

Reduced Cost

Circular
Economy



Spent grains
Wastes from breweries



Okara
Wastes from tofu makers



Molasses
Wastes from sugar refineries



Rice bran &
Sake lees
Wastes from rice processors



Ugly dates
Wastes from farms



Orange peels
Wastes from juice
manufacturers

If we can make it in Space, we can make it anywhere



Future Space Project Goals

- Independent, water-efficient protein production system
- Closed-loop, circular bioprocessing systems for foods
- Deployment in extreme environments (desert, First-Nation territories, off-grid, future space habitats)

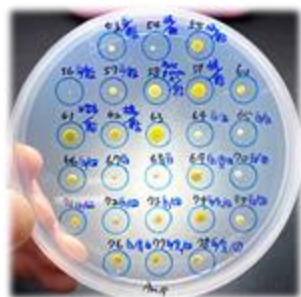


Cost Analysis vs. Other Sources of Plant-Based Proteins

	Price (€/kg)
Sophie's Bio	6.0
PEA	6.0
WHEAT	4.0
SOY – Non-GMO	4.0
SOY – GMO	2.0

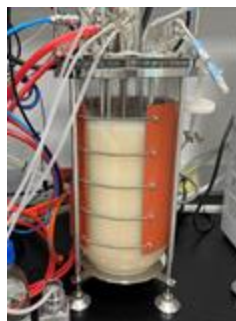
Bulk estimate – every vendor has different pricing for different customers

White strain



- Non-GMO white strain
- Resilient strain
- Multi-strains development in the pipeline for other functional proteins

Dark fermentation



Local agri-food side stream

- Low CapEx fermentation by using beer or wine brewery tanks
- Circular economy
- Zero-waste bioprocessing

Purification



Application



- MVP: Refine biomass/protein isolate/concentrate (easy to use/clean label)
- Functional properties (soluble, foaming, emulsification etc.)



Total production time under 5 days

Our strong IP Portfolio creates sustainable moat

3

Provisional patents
in the US

1

Design patents in
bioreactors

More

Trade secrets:
training microalgae
& downstream
isolation process

Target Market (B2B)



Health Ingredients &
Sports Nutrition

2025

high nutrition
gut health

US\$ 6.3 B in 2024

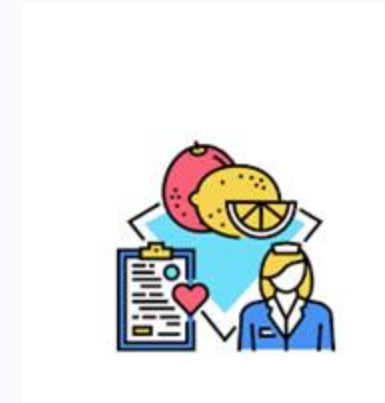


Nutraceuticals
(designer protein)

2025-2028

antioxidants,
functional peptides

US\$ 500 B in 2024



Medical Nutrition
(hydrolyzed protein)

2027-2028

complete nutrition,
naturally derived
vitamins & minerals

US\$ 51 B in 2024

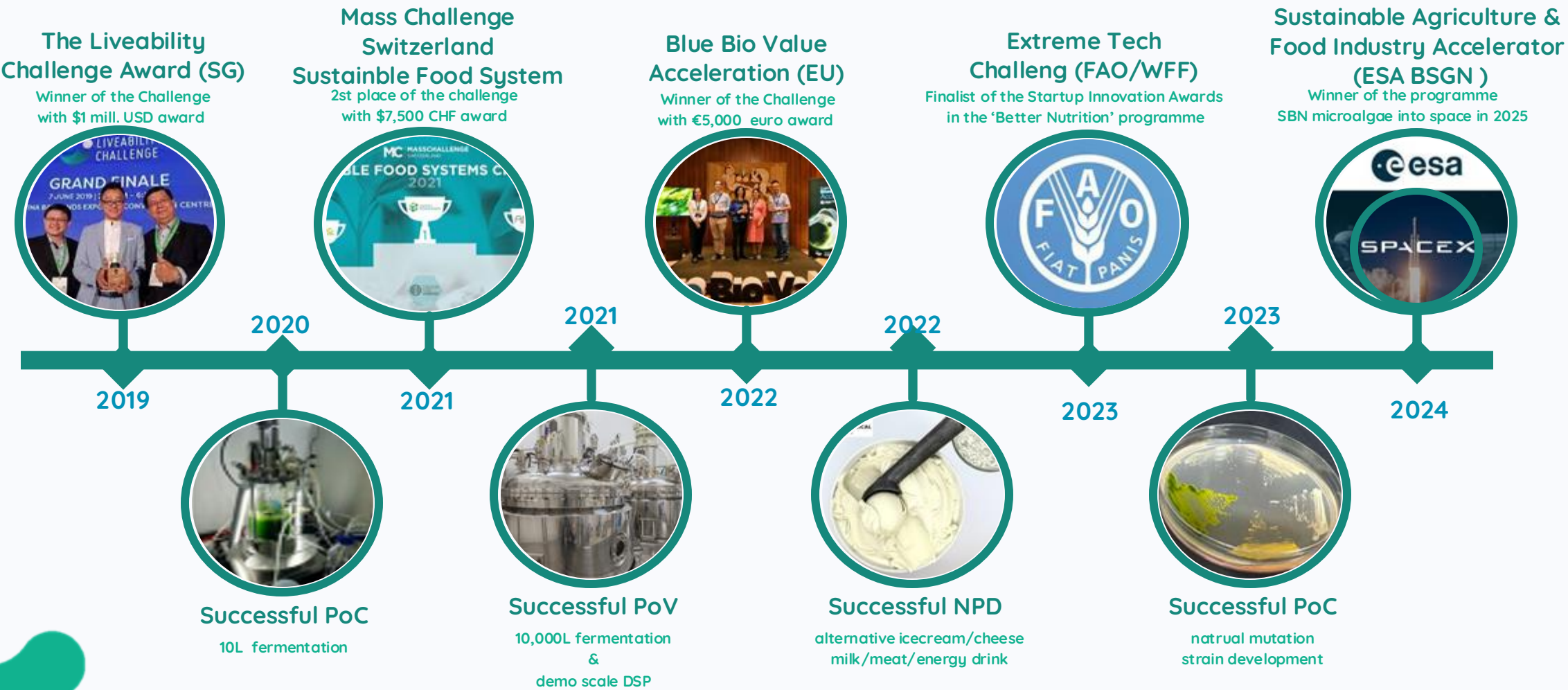


Food Ingredients
(alternative protein)

2025-2028

techno functional
protein flour for
food applications

US\$ 33 B in 2024
(dairy alternative)



Key Partners with some LOIs



R&D/ production



Commercial / validators



We are global climate responsible citizens



Eugene Wang
Co-founder &
CEO

20+ years in
food industry

Successfully
exited Sophie's
Kitchen

MBA from
Columbia
Business
School



Kirin Tsuei
Co-founder &
CTO

20+ years in
global
supplement
companies

MS in Nutrition
Science from
UC Davis

Licensed
Dietitian



Dr. Olivia Dung
COO

10+ years in
Operation
Management

Ph.D., the
Institute for Area
Studies,
Leiden University



Dr. Eric Peterson
Fermentation
Specialist

20+ years in
fermentation
research

PhD in Chem, Eng.
Queen's Univ.
Canada

Adjunct Professor
INRS
University of
Quebec



**Dr. Chen-Han
Shih**
CSO

10+ years in
microalgae
research

PhD in
Biosystems
Engineering
from The
University of
Arizona



**Dr. Chuan-
yang Fu**
Senior
Research
Fellow
10+ years in
molecular
Biology

PhD in
Molecular
Biology from
Taiwan
University



Etienne Fourquet
CFO
(part-time)

10+ years in
Finance

Worked on deals
of different sizes
internationally,
including in
Canada and the
US.



Charles Lucas
Strategy Advisor
(part-time)

10+ years in IT
Industry

Led AI projects for
industry leaders
such as Samsung
and Orange

Seeking \$2 million CAD now!!

Funding target (2026–2028)

CAD 8M

Government programs: CAD 3M

Private investment: CAD 5M

Phase 1 (2026) : lab+pilot+regulatory

- R&D lab at DRC, building core team (9 FTEs)
- Pilot process (NRC+BFT)
- Regulatory approval confirmation
- Demo-site planning & design

Phase 3 (2029)

- Volume scale-up: strategic partnership CMO
- Combined production: 1,000,000 kg
- EBITDA breakeven (+€598K)
- Expand to large multinational customers

Phase 2 (2027-2028)

- Demo-scale production facility: brownfield retrofit
- Production ramp: 50 → 100 tonnes
- First commercial sales — target EU & NA customers
- Product validation & offtake agreements

Phase 4 (2030-2031)

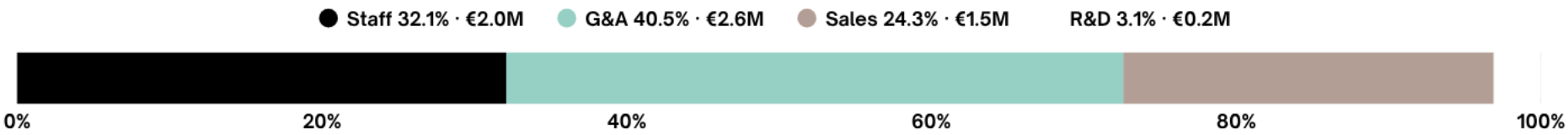
- Combined production: 3,000 → 4,500 tonnes
- Revenue €85.8M, EBITDA margin 15.8%
- Cash flow positive (2031)
- Licensing model exploration for regional expansion

EBITDA breakeven 2029 (+€598K)
Cash flow positive 2031
Revenue €85.8M / CAD \$132M (2031E)

In EUR – FYE	2026E	2027E	2028E	2029E	2030E	2031E
Revenue	—	1.2M	2.4M	21.9M	61.9M	85.8M
Gross margin	—	325K	463K	3.6M	9.5M	19.9M
Gross margin %		26.5%	19.5%	16.6%	15.4%	23.2%
OPEX	(952K)	(1.49M)	(1.36M)	(3.0M)	(4.9M)	(6.4M)
EBITDA	(952K)	(1.16M)	(896K)	+598K	+4.6M	+13.5M
EBITDA margin %		(94.8%)	(37.7%)	2.7%	7.5%	15.8%
Cumulative CF	(1.4M)	(3.7M)	(4.9M)	(5.8M)	(3.9M)	+4.5M

IRR: 17.4% | Payback: 6 years
Team: 9 → 40 FTEs
Tax: 27% blended (CCPC, PEI)
Demo-site: CAD 2M total

OPEX cost structure — €6.4M



Demo-site CAPEX — €1.4M



Let's make sustainable protein without destroying the planet

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